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EC466: Estimation and Inference in Econometrics

Exercise # 4- Answer Key

The following equation describes the median housing price in a community in terms of *lotsize* (the size of lot in square feet), *sqrft* (size of house in square feet), and *bdrms* (the number of bedrooms):

$$\begin{aligned} price_i = &\beta_0 + \beta_1 lotsize_i + \beta_2 sqrft_i + \beta_3 bdrms_i + \beta_4 lotsize^2 + \beta_5 sqrft_i^2 \\ &+ \beta_6 bdrms_i^2 + \beta_7 lotsize_i * sqrft_i + \beta_8 lotsize_i * bdrms_i \\ &+ \beta_9 sqrft_i * bdrms_i + u_i \end{aligned} \tag{1}$$

where β_j $j = 0, 1, 2, \ldots, 9$ are the true parameter values and u_i is the error term. We assume that $E(u_i|lotsize_i, sqrft_i, bdrms_i, \ldots) = 0$ and $Var(u_i|lotsize_i, sqrft_i, bdrms_i, \ldots) = \sigma^2$ for all $i = 1, 2, \ldots, n$ where σ^2 is an unknown constant and n is the number of observations. I used Gretl to obtain the regression outputs below. I estimated the standard errors of the coefficients in Model 1 under the assumption of homoskedasticity ($Var(u_i|lotsize_i, sqrft_i, bdrms_i, \ldots) = \sigma^2$ for all $i = 1, 2, \ldots, n$).

Model 1: OLS, using observations 1–88
Dependent variable: price

	Coefficient	Std. Error	t-ratio	p-value
const	15626.2	11369.4	1.374	0.1733
lotsize	−1.85951	0.637097	−2.919	0.0046
sqrft	−2.67392	8.66218	−0.3087	0.7584
bdrms	−1982.84	5438.48	−0.3646	0.7164
sqlotsize	−4.97839e–007	4.63115e–006	−0.1075	0.9147
sqsqrft	0.000352255	0.00183960	0.1915	0.8486
sqbdrms	289.754	758.830	0.3818	0.7036
lotsize_sqrft	0.000456778	0.000276890	1.650	0.1030
lotsize_bdrms	0.314647	0.252094	1.248	0.2157
sqrft_bdrms	−1.02086	1.66715	−0.6123	0.5421

Mean dependent var	3417.316	S.D. dependent var	7094.384
Sum squared resid	2.70e+09	S.E. of regression	5883.814
R^2	0.383314	Adjusted R^2	0.312158
$F(9, 78)$	5.386953	P-value(F)	0.000010
Log-likelihood	−883.3955	Akaike criterion	1786.791
Schwarz criterion	1811.564	Hannan–Quinn	1796.772

1. Report the variables that are significant at 5% level of significance.

   Answer: lotsize.

2. Test the joint significance hypothesis at 1% level of significance.

   Answer: The joint significance test implies that

$$R = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

   and $r = [0, 0, 0, 0, 0, 0, 0, 0, 0]'$. The F-statistic is 5.38 (printed in the regression output) and the critical value is 2.72. Therefore, we reject the null hypothesis at 1% level of significance.

3. Explain how you would test the following hypothesis
   $H_0 : \beta_2 = 0,\ \beta_2 + \beta_5 = 1$ and $\beta_7 = \beta_8 = 3$.

   Write down the null and alternative hypothesis in the $R\beta = r$ format where β is 10×1 parameter vector. The F-statistic should be compared with the critical value coming from $F(4, 78)$ distribution, which is 3.65 at 1% level of significance. We would reject the null at 1% level of significance if F-statistic is larger that 3.65.

Answer:

$$H_0 : R\beta = r$$
$$H_1 : R\beta \neq r$$

where

$$R = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

and $r = [0, 1, 3, 3]'$.